

TPSE II

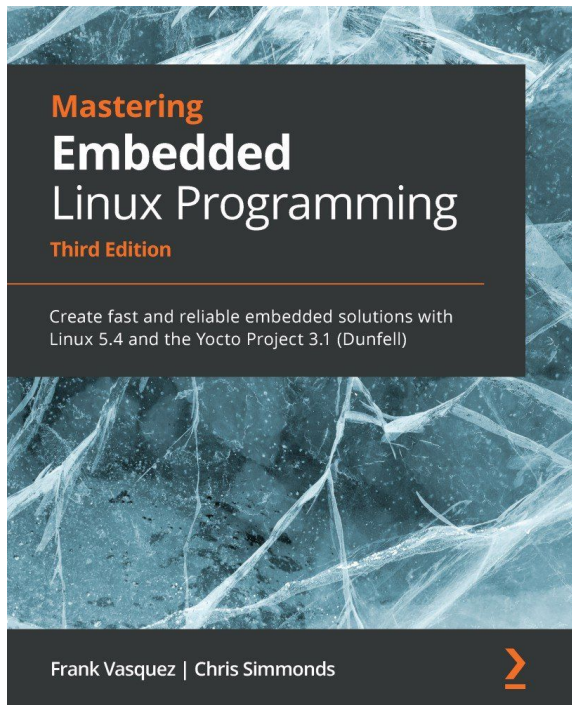
Cristiano Bacelar de Oliveira

Conteúdo

- BeagleBone Black
- Abordagens *bare metal* e com S.O.
- Toolchain
- Bootloader
- Kernel
- File System
- Comunicação SPI e I2C
- Device Drivers



Livro recomendado



Links úteis

<https://eewiki.net/display/linuxonarm/BeagleBone+Black>

http://processors.wiki.ti.com/index.php/Linux_Core_U-Boot_User's_Guide

<https://www.kernel.org/doc/Documentation/gpio/sysfs.txt>

http://processors.wiki.ti.com/index.php/Sitara_AM335x_Portal

<http://free-electrons.com/doc/training/buildroot/buildroot-labs.pdf>

<https://github.com/u-boot/u-boot/tree/master/board/ti/am335x>

<https://merongsys.wordpress.com/2015/06/29/u-boot-adding-a-new-command/>

http://wiki.beyondlogic.org/index.php?title=BeagleBoneBlack_Upgrading_uBoot

<https://releases.linaro.org/components/toolchain/binaries/5.3-2016.05/arm-linux-gnueabi/f/>



BeagleBone Black

Sintaxe dos comandos

Prompt local (no PC)

(host)\$

Prompt remoto (na BBB)

(bbb)\$



SSH

user/password

debian/temppwd

Acceso remoto (com senha)

(host)\$ ssh debian@192.168.7.2

Acceso remoto (com chave pública)

(host)\$ ssh-keygen -t rsa -b 2048

(host)\$ ssh-copy-id debian@192.168.7.2



SSH

Accesso remoto (como root)

```
(host)$ ssh root@192.168.7.2
```



SSH

scp

```
(host)$ scp <origem> <destino>
```

Copiar arquivos para a BBB

```
(host)$ scp <caminho local> debian@192.168.7.2:<caminho remoto>
```

Copiar arquivos da BBB

```
(host)$ scp debian@192.168.7.2:<caminho remoto> <caminho local>
```



TFTP Server

Instalação

```
(host)$ sudo apt-get install tftpd-hpa
```

Edite o arquivo /etc/default/tftpd-hpa

```
(host)$ sudo gedit /etc/default/tftpd-hpa
```

Altere as opções para:

```
TFTP_USERNAME="tftp"
```

```
TFTP_DIRECTORY="/srv/tftp"
```

```
TFTP_ADDRESS="0.0.0.0:69"
```

```
TFTP_OPTIONS="--secure --create --listen --verbose"
```

```
RUN_DAEMON="yes"
```



TFTP Server

Reiniciar o serviço

```
(host)$ sudo service tftpd-hpa restart
```

Verificar status

```
(host)$ sudo service tftpd-hpa status
```

Iniciar ou Interromper o serviço

```
(host)$ sudo service tftpd-hpa start
```

```
(host)$ sudo service tftpd-hpa stop
```



TFTP Client

Instalação

```
(host)$ sudo apt-get install tftp
```

Teste

```
(host)$ tftp
```

```
tftp> connect 127.0.0.1
```

```
tftp> put <file>
```

```
tftp> get <file>
```



Comunicação Serial

minicom

```
(host)$ sudo apt-get install minicom
```

```
(host)$ sudo minicom -s
```

```
(host)$ sudo minicom
```



Comunicação Serial

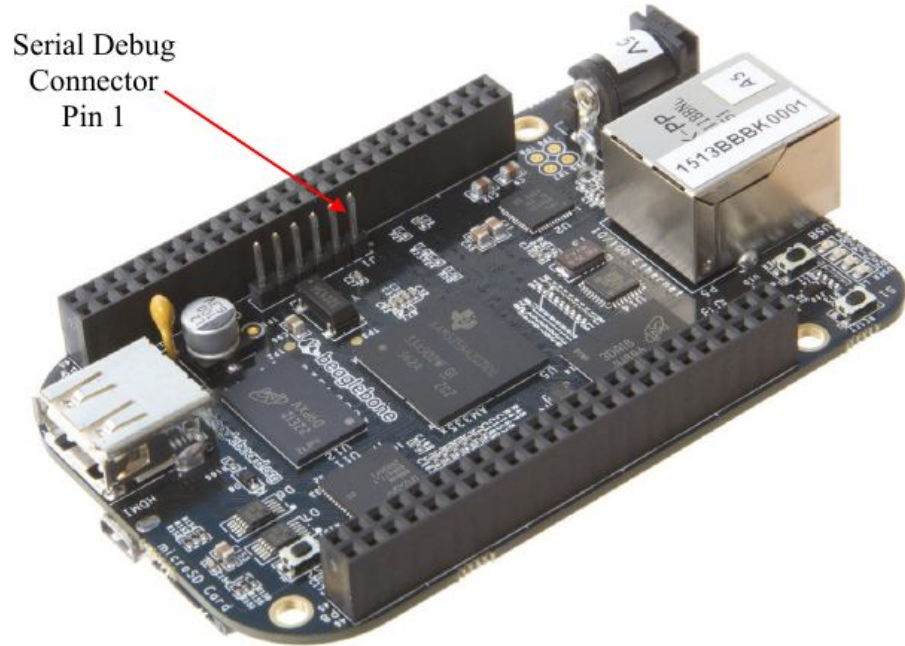
picocom

```
(host)$ sudo apt-get install picocom
```

```
(host)$ picocom <device> -b <baudrate>
```



Comunicação Serial



Comunicação Serial

Serial Debug
Connector
Pin 1

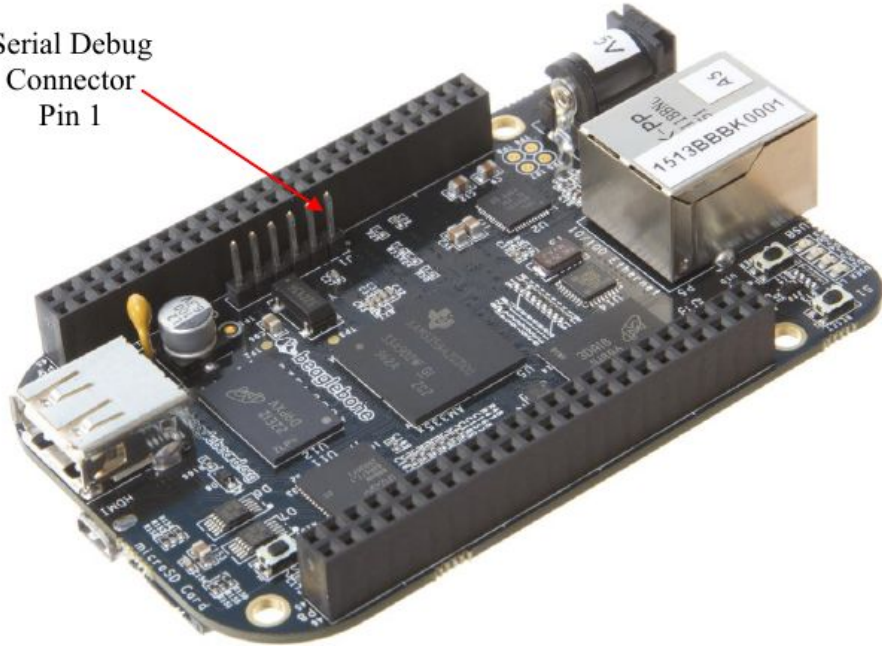


Table 14. J1 Serial Header Pins

PIN NUMBER	SIGNAL
1	Ground
4	Receive
5	Transmit

Programando na BBB

Hello World!

```
(bbb)$ vi hello.c
```

```
#include <stdio.h>
```

```
void main(void) { printf("Hello World!"); }
```

```
(bbb)$ gcc hello.c -o hello
```

```
(bbb)$ ./hello
```



Periféricos

/sys/class

```
debian@beaglebone:~$ ls /sys/class/
arvo      firmware  isku      mdio_bus  ptp       scsi_disk  ubi        watchdog
backlight gpio      kone      mem        pwm       scsi_host  udc
bdi       graphics  koneplus  misc       pyra      sound      uio
block     hidraw    kovaplus  mmc_host   rc        spi_master usbmon
bsg       hwmon     lcd        mtd        regulator spidev     vc
dma       i2c-adapter leds       net        rtc        thermal    video4linux
drm       i2c-dev   logibone  power_supply savu      timed_output virtio-ports
dvb       input     mbox      pps        scsi_device tty        vtconsole
```

GPIO

4 GPIO Ports

- GPIO0, GPIO1, GPIO2, GPIO3
 - gpiochip0, gpiochip32, gpiochip64, gpiochip96
 - Ex. GPIO2[10] corresponde ao índice 74

Usando o driver de GPIO

```
(bbb)$ echo 60 > /sys/class/gpio/export
```

```
(bbb)$ echo out > /sys/class/gpio/gpio60/direction
```

```
(bbb)$ echo 1 > /sys/class/gpio/gpio60/value
```



Bare Metal Embedded Systems

Cross-Compile

Instalação

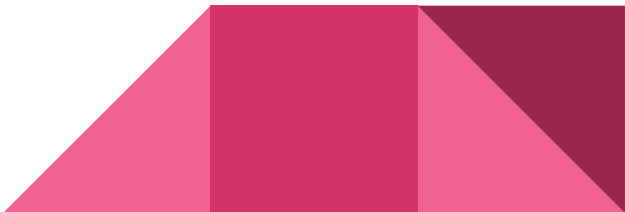
```
(host)$ sudo apt-get install gcc make
```

```
(host)$ sudo apt-get install gcc-arm-linux-gnueabi
```

```
(host)$ sudo apt-get install binutils-arm-linux-gnueabi
```

Teste

```
(host)$ arm-linux-gnueabi-gcc --version
```



Sistemas Operacionais

Toolchain

Bootloader

Linux Kernel

File System



SPI: Serial Peripheral Interface



I2C: Inter-Integrated Circuit

Linux Device Drivers